

SIR Epidemic Model

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Dynamical Systems / ODEs

Universe

- Population $N = 10000$
- Time horizon: $[0, 200.0]$ days

Variables

Symbol	Meaning	Type / Range
$S(t)$	susceptible individuals at time t	$[0, 10000]$
$I(t)$	infected individuals at time t	$[0, 10000]$
$R(t)$	recovered individuals at time t	$[0, 10000]$
β	transmission rate = 0.3/day	$\mathbb{R}_+$
γ	recovery rate = 0.1/day	$\mathbb{R}_+$
R_0	basic reproduction number = $0.3/0.1$	$\mathbb{R}_+$
N	total population = 10000	$\mathbb{Z}^+$

Structure

A disease outbreak in a population of $N=10,000$. The transmission rate is $\beta=0.3/\text{day}$ and the recovery rate is $\gamma=0.1/\text{day}$ (10-day infectious period). Initially 10 individuals are infected, with the rest susceptible. Questions:

- What is the basic reproduction number R_0 ?
- When does the infection peak, and how many are infected at the peak?
- What fraction of the population is ultimately infected?
- What vaccination rate prevents an epidemic (herd immunity threshold)?

Real-World Mapping

Real-World Concept	Mathematical Object
healthy population	$S(t)$ — susceptible compartment
sick population	$I(t)$ — infected compartment
immune population	$R(t)$ — recovered compartment

contagiousness

beta — transmission rate

recovery speed

gamma — recovery rate

Constraints

(1)

$$\frac{dS}{dt} = -\frac{\beta SI}{N}$$

susceptible individuals become infected

(2)

$$\frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I$$

infected gain from transmission, lose from recovery

(3)

$$\frac{dR}{dt} = \gamma I$$

recovered individuals accumulate

(4) $S(t) + I(t) + R(t) = N$

population conservation

(5)

$$R_0 = \frac{\beta}{\gamma}$$

basic reproduction number

(6)

$$v^* = 1 - \frac{1}{R_0}$$

herd immunity threshold

Objective

$$\text{Solve } \frac{d\mathbf{y}}{dt} = f(t, \mathbf{y}), \quad \mathbf{y}(0) = \mathbf{y}_0$$

Approach

Suggested approach Runge-Kutta 4(5) adaptive ODE integration (scipy.integrate.solve_ivp)

Complexity class:

Available tools: Runge-Kutta 4

2. Solution

Answer:	$R_0 = 3.00$; peak infected = 3,008; herd immunity = 66.7%
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	Runge-Kutta 4(5) adaptive ODE integration (scipy.integrate.solve_ivp)
Time:	0.1349s
Certificate:	Population conservation verified. $S(t)+I(t)+R(t)=N$ at all time steps.

Solution Details

Population: $N = 10,000$

Transmission rate: $\beta = 0.3/\text{day}$

Recovery rate: $\gamma = 0.1/\text{day}$

Initial infected: $I(0) = 10$

Basic reproduction number: $R_0 = \beta/\gamma = 3.00$

Peak infection: $I_{\max} = 3,008$ at $t = 38.3$ days

Final epidemic size (numerical): 94.1% of population

Final epidemic size (theory): 94.0% of population

Herd immunity threshold: $v^* = 1 - 1/R_0 = 66.7\%$

Vaccination scenarios:

$v = 0\%$: $R_{\text{eff}} = 3.00$, peak = 3,015, final size = 94.2% [EPIDEMIC]

$v = 30\%$: $R_{\text{eff}} = 2.10$, peak = 1,204, final size = 57.7% [EPIDEMIC]

$v = 50\%$: $R_{\text{eff}} = 1.50$, peak = 325, final size = 28.8% [EPIDEMIC]

$v = 67\%$: $R_{\text{eff}} = 0.99$, peak = 10, final size = 1.5% [contained]

$v = 80\%$: $R_{\text{eff}} = 0.60$, peak = 10, final size = 0.2% [contained]

Disease-free equilibrium: unstable

Eigenvalues: 0.0, 0.0, 0.19999999999999998

Verification

✓ Population Conserved	Passed
✓ R_0 Correct	Passed
✓ Peak Occurs	Passed
✓ Peak Time Reasonable	Passed

- ✓ Final Size Matches Theory Passed
- ✓ Herd Immunity Correct Passed
- ✓ Vaccination Prevents Epidemic Passed
- ✓ All Compartments Nonneg Passed

3. Interpretation

The Question

A disease outbreak in a population of $N=10,000$.

The Answer

$R_0 = 3.00$; peak = 3,008 infected at day 38; herd immunity at 67% vaccination.

What This Means

- R_0 : 3.0
- Peak infection: around day ~30, with ~2,900 infected simultaneously
- Final epidemic size: ~94% of the population ultimately infected
- Herd immunity threshold: 67% vaccination rate prevents an epidemic
- Vaccination analysis: effective R_0 drops below 1.0 at 67% coverage, preventing large outbreaks

Recommendations

1. Monitor R_0 closely; small changes in transmission can shift epidemic dynamics dramatically.

Limitations

- SIR model assumes homogeneous mixing; real populations have network structure.

- Parameter time.